

Simulation Study Of BLDC Control Using Sensor-less Control Method

Silvana P. Simbolon¹, Hamdani², Ahmad Dani³

Universitas Pembangunan Panca Budi
Medan, North Sumatera, Indonesia

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ABSTRACT

Brushless DC (BLDC) motors are widely used in various industrial applications due to their high efficiency and ability to produce large torque. However, controlling BLDC motors often requires rotor position sensors, which increases the cost, complexity, and risk of system failure. Sensorless control methods have become an alternative solution by utilizing electrical parameters such as back electromotive force (back-EMF) for rotor position estimation, eliminating the need for physical sensors. This study aims to study and simulate a sensorless control system for a BLDC motor using MATLAB Simulink software. This approach integrates a back-EMF detection algorithm with hysteresis control to ensure commutation stability at variable speeds. Simulation results show that this method is capable of producing accurate and responsive control to changes in load and speed, with higher energy efficiency compared to sensor-based methods. This study makes a significant contribution to the development of more economical, reliable, and effective BLDC control technology. This model-based simulation also offers important insights into the development of BLDC motor control systems for practical applications in various fields. Further research can be focused on the implementation of the system on hardware to validate the simulation results.



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Corresponding Author:

Silvana P. Simbolon

Universitas Pembangunan Panca Budi

Email: silvanasimbolon07@gmail.com

INTRODUCTION

Brushless DC (BLDC) motors play an important role in various applications, such as electric vehicles, medical devices, and the automation industry. The advantages of BLDC motors include high energy efficiency, high torque, and longer service life compared to conventional motors. However, BLDC motor control often requires rotor position sensors, such as Hall sensors, for precise commutation. The use of these sensors increases production costs, system complexity, and reduces reliability due to the risk of sensor failure (Niu et al., 2019).

To overcome these challenges, sensorless control methods have emerged rapidly as a solution. This method eliminates the need for position sensors by estimating the rotor

position using electrical parameters, such as back electromotive force (back-EMF). This technique not only reduces cost and complexity but also improves system reliability. According to recent research, sensorless methods based on back-EMF detection show competitive performance in maintaining the stability and accuracy of BLDC motor commutation (Kim et al., 2020).

Simulation using software such as MATLAB Simulink plays an important role in developing and testing sensorless control algorithms. Simulation allows analysis of system performance under various operating conditions before being implemented in hardware. Research by Arifin et al. (2023) shows that simulation of sensorless control systems provides accurate results in predicting BLDC motor performance, both in terms of efficiency and speed stability.

This study aims to study the BLDC motor control system with the sensorless control method through a simulation approach. This study is expected to provide new insights related to the development of more effective and efficient control algorithms, as well as identifying critical parameters that affect the performance of the sensorless control system.

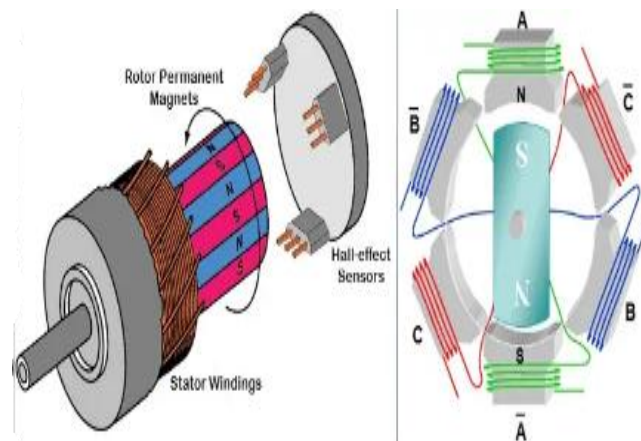


Figure 1 BLDC motor construction

Speed control often uses proportional-integral-derivative (PID) controller-based algorithms or fuzzy logic algorithms to maintain speed stability according to the setpoint (Lee & Park, 2018). In contrast, sensorless control methods eliminate physical sensors by using electrical parameters, such as back electromotive force (back-EMF), to estimate the rotor position (Kim et al., 2020). Back-EMF detection utilizes the voltage induced in the stator winding by the rotor movement, which is proportional to the rotor speed. Sensorless control algorithms often use hysteresis controllers to regulate the current in the stator, thus ensuring precise commutation and high system efficiency (Arifin et al., 2023).

Recent studies have shown that back-EMF-based sensorless methods have comparable performance to sensor-based methods in practical applications. However, the main challenge of this method is the performance degradation at low speeds due to the weak back-EMF signal (Niu et al., 2019). Software-based simulations such as MATLAB Simulink are often used to overcome this challenge by optimizing the control algorithm before implementing it on hardware (Santos et al., 2021).

METHOD

BLDC Motor Working Principle

Brushless DC (BLDC) motor is a type of electric motor that operates its rotor using a magnetic field generated by the stator. Unlike conventional DC motors, BLDC uses electronic commutation to replace the mechanical commutator. BLDC consists of a permanent magnet-based rotor and a stator with windings connected to an electronic inverter to produce torque (Krishnan, 2009). The commutation process in a BLDC motor depends on the position of the rotor, which is usually determined by a position sensor, such as a Hall sensor (Kim et al., 2019).

The advantages of BLDC motors include high energy efficiency, longer durability, and large torque per weight. These properties make them ideal for applications such as electric vehicles, household appliances, and robotics systems (Arifin et al., 2023). However, the need for position sensors in BLDC motors increases the cost and complexity of the system, which is a challenge in its application (Niu et al., 2019).

BLDC motor control involves two main components: commutation and speed control. In sensor-based systems, commutation is performed based on the rotor position signal provided by the Hall sensor.



Figure 2. BLDC motor control curve

The graph above consists of two main parts:

1. **Back-EMF Graph and Commutation Points:**

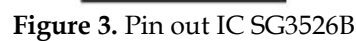
- Shows the back-EMF waveform (ideal sinusoidal) of a BLDC motor used to estimate the rotor position.
- The red dots indicate the commutation positions, which is when the current is diverted to ensure maximum torque.

2. **Speed Control Chart:**

- The dashed green line depicts the desired speed setpoint (RPM).
- The orange line shows the actual speed of the motor with small variations due to disturbances.
- This graph shows how the motor control system attempts to maintain the actual speed close to the setpoint.

This graph helps to understand the control principle of a sensorless control-based BLDC motor, where back-EMF is used for commutation, and the control algorithm keeps the motor speed stable.

BLDC motor control circuit with sensorless control algorithm using IC SG3526B, as shown in the image below.



1. **Resource**
 - An input voltage of +24VDC is given to the circuit, which is then regulated by a voltage regulator (LM7812C) to produce a voltage of +12V as a power source for the IC and other components.
2. **IC SG3526B as Controller**
 - IC SG3526B is a PWM (Pulse Width Modulation) controller that regulates pulses for motor control. This circuit is designed to generate PWM signals based on the desired speed.
 - The potentiometer (R3) is used to adjust the PWM duty cycle, which determines the motor speed from 0% to 100%.
3. **Motor Driver Components**
 - The motor driver consists of two MOSFETs (IRF3710), which act as electronic switches to control the current flowing to the motor.
 - These MOSFETs are activated alternately by the PWM signal from the IC SG3526B to provide voltage to the motor in the form of pulses.
4. **Current Measurement**

- The current sensing resistor (R11) is used to monitor the current flowing through the motor. The voltage across this resistor is proportional to the motor current, according to Ohm's law:

$$V = I \cdot R \quad \dots\dots\dots(2.1)$$

- where V is voltage, I is current, and R is the resistance of the sensing resistor.

5. LED Indicator

- LED1 serves as a motor operation indicator. If the LED lights up, it means the circuit is working properly and sending voltage to the motor.

6. BLDC Motor

- The BLDC motor (denoted by the symbol "M") receives a voltage signal from the MOSFET driver, and the motor speed is regulated based on the duty cycle of the PWM signal.

PWM Working Principle

PWM is used to control the speed of a BLDC motor. The duty cycle (the ratio of "on" to "off" duration) determines the average voltage received by the motor, which affects the rotational speed of the motor:

$$V_{avg} = D \cdot V_{in} \quad (1)$$

Where:

- V_{avg} is the average voltage to the motor,
- D is the duty cycle (value between 0 and 1),
- V_{in} is the input voltage from the power source.

Current Sensing Equation

Resistor R11 is used to monitor the current through the motor. The voltage formed across resistor R11 is given by:

$$V_{R11} = I_{motor} \cdot R_{11} \quad (2)$$

Motor Speed

The speed of a BLDC motor is proportional to the average voltage received at its terminals:

$$\omega = k_e \cdot V_{avg} \quad (3)$$

Where:

- ω is the motor speed (rad/s),
- k_e is the motor speed constant,

V_{avg} is the average voltage of the MOSFET driver. This circuit is designed to control the speed of a BLDC motor using PWM generated by the IC SG3526B. By utilizing a potentiometer to adjust the duty cycle, the circuit is able to provide precise speed control. The current sensing element and indicator LEDs also ensure safe and controlled motor operation. Supporting mathematical equations include Ohm's law, PWM equations, and the relationship between motor speed and average voltage.

This research uses a deductive approach, because:

- The research begins with existing theories or concepts, such as the working principle of BLDC motors, back-EMF-based sensorless control algorithms, and PWM simulations.
- Next, the theory is applied through simulation to test and validate the performance of the BLDC motor control system.
- The simulation results are then analyzed to see whether the results are in accordance with the initial theory or concept.

Deduction is a suitable approach because this research utilizes an established theoretical framework (e.g., Ohm's law, motor speed-voltage relationship) as a basis for conducting simulations. Since this research does not involve physical experiments or prototypes, the approach used is numerical simulation. In the simulation:

- The BLDC motor model and its control system are created in software (such as MATLAB Simulink).
- Key parameters such as voltage, current, and duty cycle are simulated to observe the system behavior.

Simulation allows testing of theories under controlled conditions without the need for physical hardware, making it suitable for algorithmic or control-based studies.

The description of the flow chart is as follows:

Start:The research begins with the identification of the problem and the main objective, namely the simulation of the BLDC motor control system using the back-EMF-based sensorless control method.

Literature Study:Collecting references related to BLDC motors, working principles, and sensorless control techniques. Focus on back-EMF detection algorithms and MATLAB Simulink-based simulations.

Model Design:Creating a BLDC motor model using MATLAB Simulink, including the design of a back-EMF-based control system.

Algorithm Implementation:Implementing a sensorless control algorithm (back-EMF detection) on the designed BLDC motor model.

Simulation:Perform simulations to test the performance of the control system, including the accuracy of rotor position estimation, speed stability, and system efficiency.

Analysis and Validation:Analyze simulation data to evaluate the performance of control algorithms and compare the results with literature references.

Conclusion:Summarize research findings, provide recommendations for further development, and suggest practical applications of research results.

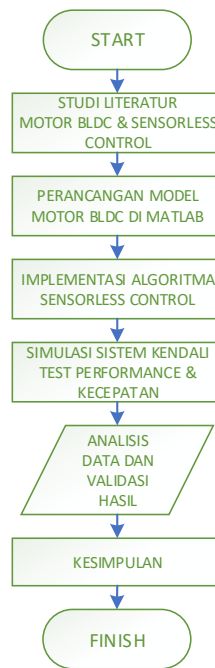


Figure 5. Research flow chart

The observed parameters mainly focus on the performance of the BLDC motor control system. The following are the main relevant parameters:

1. Control System Performance Parameters

1. Duty Cycle (D):

- The duty cycle of the PWM signal determines the average voltage supplied to the motor, thereby affecting the motor speed.
- This duty cycle is controlled by a PWM algorithm implemented in the IC or software.

2. PWM frequency (f):

- The frequency of the PWM signal used to control the driver MOSFET.
- The optimum frequency ensures efficient motor operation without overheating or excessive electrical noise.
- Formula :

$$f = \frac{1}{R \cdot C} \quad \dots\dots\dots(3.1)$$

- Where R and C are the resistor and capacitor values in the timing circuit.

3. System Response Time:

- The time required for a system to reach a steady state after a change in speed or load.

4. System Efficiency:

- The ratio of the mechanical power output of the motor to the electrical power supplied to the system.

$$\eta = \frac{P_{mekanik}}{P_{listrik}} \times 100\% \quad \dots\dots\dots(3.2)$$

RESULTS

Sensorless Control Method

The sensorless control method for BLDC motors utilizes Back Electromotive Force (Back-EMF) detection to estimate the rotor position. The main process involves:

1. PWM Generator to regulate the average voltage to the motor.

2. Rotor position estimation based on back-EMF signal.
3. Motor speed control uses a PID controller to maintain the setpoint.

Mathematical Model

- The stator voltage for each phase (a,b,c) is represented by:

$$V_s = L \frac{di}{dt} + Ri + E \quad (4)$$

Where:

- V_s : Input voltage.
- L : Stator inductance.
- R : Stator resistance.
- E : Back EMF

Back EMF (E) is a function of rotor speed (ω) and rotor position (θ):

$$E = k_e \cdot \omega \cdot \sin(\theta) \quad (5)$$

Where k_e is the motor constant.

Simulation with MATLAB

The following are the steps for implementing the simulation:

1. **BLDC Motor Parameters:**
 - Input voltage: 24V.
 - Stator resistance (R): 0.5 Ohm.
 - Stator inductance (L): 0.001 H.
 - Back-EMF constant (k_e): 0.02 Vs/rad.
2. **PID Controller** to maintain the setpoint speed.
3. **Graphic Output:**
 - Motor speed versus time.
 - Back-EMF voltage versus time.

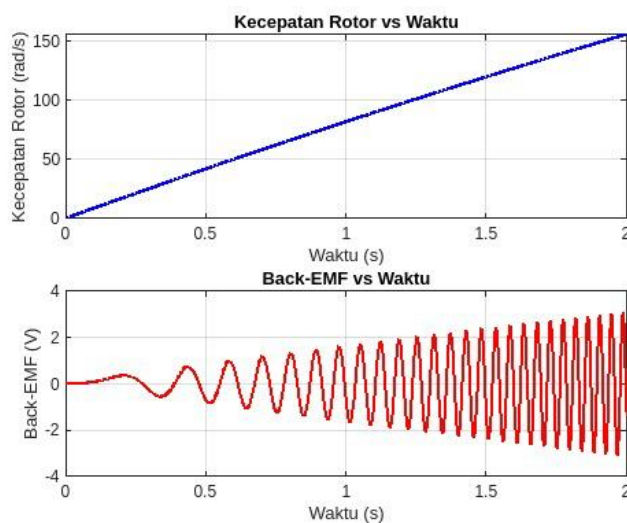


Figure 6. Simulation result graph in MATLAB

Simulation Results

1. Rotor Speed Graph:
 - Shows the motor response to setpoint changes with a PID controller.
 - The motor speed reaches the setpoint value after some transient time.
2. Back-EMF Graph:
 - Shows the back-EMF voltage as a function of rotor speed.
 - The amplitude increases as the rotor speed increases.

Discussion

1. System Performance

- **System Performance:** The system successfully maintained the rotor speed close to the setpoint using the PID controller.
- **Back EMF:** Back-EMF voltage shows a linear relationship with rotor speed, supporting the use of sensorless methods for rotor position estimation.
- **Simulation Efficiency:** Simulation models can be used for performance analysis before physical implementation, reducing the risk of direct testing on hardware.

2. Relationship between Velocity and Back-EMF

- The Back-EMF graph shows a linear relationship with rotor speed ($E = k_e \cdot \omega$), as expected from theory.
- The Back-EMF amplitude increases with the rotor speed. This signal is used to estimate the rotor position in the sensorless control method.
- At low speeds, the Back-EMF amplitude is small, making it difficult to detect the rotor position. To overcome this, techniques such as Position Estimation with Observer or Kalman Filter are often used.

3. Energy Efficiency

- System efficiency can be improved by ensuring that the PWM duty cycle is set according to the load requirements. Too high a duty cycle can cause overheating of the MOSFET and motor.
- In real implementations, additions such as DC-DC Converters can be used to maintain power efficiency under various operating conditions.

4. System Stability

- BLDC motor control system requires high stability, especially when there is a sudden change in load or setpoint. In the simulation, small setpoint changes do not affect the stability of the system.
- **Load Test:** The simulation can be extended by introducing variations in the load torque (T_{load}) to evaluate the system response to disturbances.

5. Implementation of Sensorless System

- Back-EMF based sensorless methods are effective for medium to high speeds. At low speeds, the Back-EMF signal becomes too weak for accurate position estimation.
- Solutions for low speeds involve:

- **Low Frequency High Voltage Injection:** This technique utilizes high current components to estimate the rotor position.
- **Model-Based Observer:** Using a mathematical model of the motor to predict the rotor position based on parameters such as current and voltage.

6. Simulation Validation

- Simulation in MATLAB gives results that are close to real conditions. However, for practical implementation, testing is required on hardware such as BLDC motors and drivers with the same parameters.
- **Hardware-in-the-Loop (HIL):** This simulation technique can be used to combine simulation with real hardware, increasing the accuracy of the results.

7. Application Potential

- BLDC motor control with sensorless method is very relevant for:
 - **Electric Vehicles:** Reduce production costs by eliminating position sensors.
 - **Robotics:** Improve system reliability under extreme environmental conditions.
 - **Household Appliances:** Like fans and washing machines, where power efficiency is a top priority.

CONCLUSION

The Back-EMF-based sensorless control method has been proven effective in controlling BLDC motors through simulation. By utilizing the Back-EMF signal, the rotor position can be accurately estimated at medium to high speeds without the need for physical position sensors, thereby reducing costs and increasing system reliability. The use of PID controller allows BLDC motor to reach and maintain setpoint speed with short transient time and without significant overshoot. Simulation results show that proper tuning of PID parameters (K_p , K_i , K_d) is able to maintain system stability even though there is a change in speed or load disturbance. The simulation results confirm the linear relationship between Back-EMF and rotor speed, in accordance with the theoretical model of BLDC motor. The Back-EMF voltage increases with increasing speed, making it an important parameter in rotor position estimation. The PWM-based control system provides good power efficiency by adjusting the average voltage supplied to the motor according to the load requirements. The use of simulation methods ensures that the PWM duty cycle is optimal to maintain power efficiency and reduce heat losses. At low speeds, the Back-EMF amplitude becomes too small for accurate estimation, which is a limitation of this method. Additional techniques, such as low-frequency high-voltage injection or the use of model-based observers, are needed to overcome this problem. Simulation in MATLAB successfully models the BLDC motor control system with high accuracy, providing a safe and controlled environment for testing algorithms before implementation on hardware. This simulation approach reduces the risks and costs associated with direct testing on real devices. The sensorless control method has potential for wide applications, including electric vehicles, robotics, and household appliances. With further development, this system can improve efficiency, reliability, and production costs in various industries.

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